🧠 InsightBoard

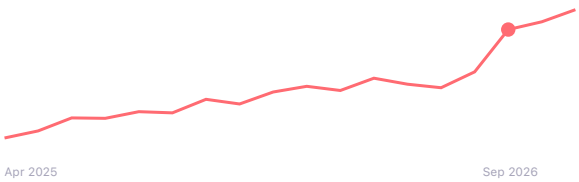
LLM-powered decision intelligence for a Bharti Airtel–style telecom: which of the many KPIs actually explain revenue and margin? (Synthetic illustrative data, 18 months to Sep 2026.)

Explain: Revenue growth (%) **in month:** Jul 2026

Actual change	vs average month	Explained by drivers
+4.26%	+3.53%	+3.38%

Model fit (R^2)
0.88

Monthly revenue (₹ crore) – Jul 2026 highlighted



Why did it change? (driver contributions)

Premium plan share % (lag 0)	+1.10%
Digital recharge share % (lag 0)	+0.91%
Network uptime % (lag 1)	+0.47%
Training hours (lag 0)	+0.40%
Customer churn % (lag 2)	+0.28%
Support utilization % (lag 1)	+0.21%

Unexplained: +0.15%

KPI evidence – 14 indicators tested

KPI	Dept	Raw r (levels)	Change r	Best lag	Ver
Premium plan share %	Commercial	0.98	0.69	0 mo	Dr

Digital recharge share %	Commercial	0.88	-0.66	0 mo	Dr
Training hours	People	0.89	0.66	0 mo	Dr
Support utilization %	People	-0.00	-0.64	1 mo	Dr
Customer churn %	Customer	0.21	0.64	2 mo	Dr
Network uptime %	Network	-0.06	0.57	1 mo	Dr
Complaint resolution h	Customer	0.05	-0.55	1 mo	Dr
Call drop rate %	Network	0.03	-0.54	1 mo	Dr
Avg data speed Mbps	Network	0.68	0.51	0 mo	Dr
NPS	Customer	-0.00	0.50	0 mo	Dr
5G adoption %	Network	0.93	0.49	0 mo	Dr
Store footfall idx	Commercial	-0.78	0.37	0 mo	Sp
Attrition %	People	0.02	0.33	0 mo	No
Marketing spend ₹cr	Commercial	0.20	-0.13	2 mo	No

Executive brief (LLM)

Generate executive brief

The LLM is given only the computed evidence below, nothing else.

Click to generate. Without LLM access the page shows a rule-based summary.

Ask e.g. Why did revenue growth drop in May 2026?

Ask

Answers are grounded in the evidence table.

Algorithm used by the system

1. **Ingest** 14 operational KPIs (network, customer, commercial, people) and two financial

outcomes (revenue, EBITDA margin).

2. **Remove shared trends**: use month-over-month changes. Two series that merely both rise over time have a high raw correlation but a near-zero correlation in changes.
3. **Lag search**: correlate each KPI change with the outcome change at lag 0, 1 and 2 months and keep the best. Significance threshold `|r| > 2/√n`.
4. **Classify** each KPI as *Driver* (significant), *Spurious trend* (high raw r, no change-r) or *Noise*.
5. **Joint model**: ridge regression on standardized, lag-aligned changes of the top drivers (λ = 1.5), to separate overlapping signals.
6. **Attribute a month**: contribution = β × (that month's standardized KPI change) × outcome spread; the rest is "unexplained".
7. **Evidence pack → LLM**: top drivers with r, lag, contribution and KPI movement are sent as JSON; the LLM writes the headline, insights, actions and caveats using only that evidence.
8. **Guardrail**: correlation is not proof of causation; the brief must say so and recommend a validation step.